

Can a Paper’s Title Predict Its Citations?

Sorting Computer-Science Papers into Citation Bands, and Finding Out Which Information Actually Helps

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Abstract—When a computer-science paper is published, what information available at that moment predicts how often it will be cited? We frame this as a four-class classification problem: sorting papers into citation bands (Uncited, Low, Medium, High) based on citations received in a fixed two-year window after publication, using a dataset of 43,997 papers pulled from OpenAlex (2019–2023). Starting from a CS229 (Fall 2019) project that predicted citation bins from title, journal and year and observed scores plateau regardless of model size, we test directly whether the ceiling comes from the features rather than the model. Using multinomial logistic regression with a controlled feature ablation (title text, metadata, venue), we find that titles alone are weak predictors (macro-F1 0.361, vs. a 0.105 majority-class baseline), metadata and venue each carry more signal than the title, and the combined feature set reaches 0.450 macro-F1 on held-out test data, a ceiling that larger models (Random Forest, HistGradientBoosting) do not break.

Index Terms—citation prediction, text classification, feature ablation, scholarly metadata, logistic regression

I. INTRODUCTION

When a computer-science paper comes out, what do we know that says how often it will be cited? We sort each paper into one of four bands, from no citations at all to eight or more in its first two years. This is a plain multi-class classification task; it has nothing to do with citation networks or with why one paper cites another, even though the problem title can be read that way.

We started from a CS229 project (Fall 2019) that predicted citation bins from a paper’s title, journal and year [1]. Its authors observed that scores stopped improving regardless of how large the network grew, and suggested that the input features were the limiting factor, but did not test this directly. That became our question: if a bigger model does not help, is it the information we feed it?

We also changed four things relative to the original setup. First, the original predicted raw citation counts, which depend on how long a paper has been out; we count citations inside a fixed window instead. Second, its bin edges were taken from the training data; ours are fixed and explained. Third, its report compares models against each other; ours also shows a majority-class baseline next to every score. Fourth, its journal was encoded as a single numeric code; ours is one-hot encoded and tested in isolation.

II. DATASET

No dataset came with the problem statement, so we built one from the OpenAlex API [2]. We took English-language

works of OpenAlex type `article` in Computer Science published from 2019 to 2023 that have a title and at least one reference, and drew 800 papers at random for every year and CS subfield combination ($5 \text{ years} \times 11 \text{ subfields} = 55 \text{ strata}$, fixed seed 42). This gives **43,997 papers**, each with its title, year, subfield, topic, venue, number of references, and citation counts. The raw (lifetime) citation total is kept for comparison but never used as a feature. Only the venue field is ever missing (602 papers, 1.4%). Papers from 2024 are excluded because their second citation year is not yet complete.

The dataset is published as a Kaggle dataset named `tharunganesh172/citations-openalex-cs-2019-2023`. We split it once, 70/15/15 into train, validation and test (30,797 / 6,600 / 6,600 papers, balanced by band and year), and saved the split so that every experiment sees exactly the same papers.

III. TARGET DEFINITION AND CLASS BANDS

Total lifetime citations favor older papers, so we count only the citations a paper receives in the two calendar years following its publication year. Every paper is measured over an identical two-year window, which removes paper age as a confound by construction rather than correcting for it statistically.

The citation counts are strongly right-skewed: 26.5% of papers receive zero citations in the window, the median is 2, and the 99th percentile is 64. We cut the counts into four bands at approximately the quartiles: **0**, **1–2**, **3–7**, and **8 or more** citations, labeled Uncited, Low, Medium and High. Each band holds roughly a quarter of the papers, so a majority-class baseline scores only 0.265 accuracy and 0.105 macro-F1, with random guessing at 0.25 accuracy.

Fig. 1 is our validation that the window construction works: if the target is genuinely age-invariant, papers from different publication years should look alike. They do: neither the mean citation count nor the fraction of uncited papers shows a trend across 2019–2023.

IV. METHOD

We keep the model family fixed and vary only what it is allowed to see. Every primary result in this paper comes from multinomial logistic regression. Features are produced by one shared pipeline with three blocks that can be independently switched on or off:

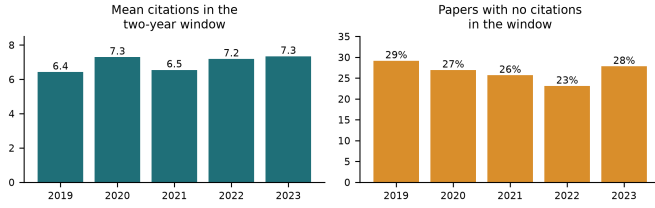


Fig. 1. Papers from every publication year look alike under the two-year window, confirming that paper age no longer drives the target.

TABLE I

MACRO-F1 FROM CROSS-VALIDATION AND ON THE HELD-OUT TEST SET. THE TWO AGREE TO WITHIN ABOUT 0.01 (LARGEST GAP 0.011, FOR THE TITLE-ONLY MODEL), CONFIRMING NOTHING WAS TUNED ON THE TEST SET.

Feature set	5-fold CV	Test F1	Test acc.
Majority-class baseline	n/a	0.105	0.265
Title only	0.372	0.361	0.368
Venue only	0.385	0.390	0.420
Title + metadata	0.424	0.420	0.429
Metadata only	0.427	0.423	0.443
Title + metadata + venue	0.448	0.450	0.459

- **Title:** word-level TF-IDF (1–2 grams, $\text{min_df} = 3$, up to 50,000 features, sublinear term frequency) concatenated with character-level TF-IDF (3–5 character n-grams on word boundaries, $\text{min_df} = 5$, up to 50,000 features);
- **Metadata:** one-hot encoded subfield and topic (rare categories below a frequency of 5 are grouped into an “infrequent” bucket), plus publication year and $\log(1 + n_{\text{references}})$, standardized;
- **Venue:** one-hot encoded venue name, with the same rare-category grouping as the metadata block.

Each feature combination was evaluated with 5-fold cross-validation on the 37,397 training-plus-validation papers, then scored once on the held-out 6,600 test papers. The logistic regression classifier uses L_2 regularization ($C = 1.0$) with `class_weight="balanced"`, since one band otherwise dominates the loss. Our primary metric is macro-F1, the unweighted average of the four per-band F1 scores, which cannot be inflated by always predicting the majority band. All combinations used the same split, the same evaluation code, and a shared results log; code is covered by automated tests (pytest). A small web demo accepts a title and returns the predicted band, class probabilities, and the words that drove the prediction; it uses only the title-only model.

V. RESULTS

Table I summarizes the ablation. Four findings stand out.

1) *The title helps, but only a little.* It beats the baseline clearly (0.361 vs. 0.105), yet it is the weakest individual source of signal. Once metadata is known, the title adds essentially nothing: 0.423 without the title versus 0.420 with it.

2) *Metadata and venue carry more signal.* In early exploration, the number of references alone reached about 0.40

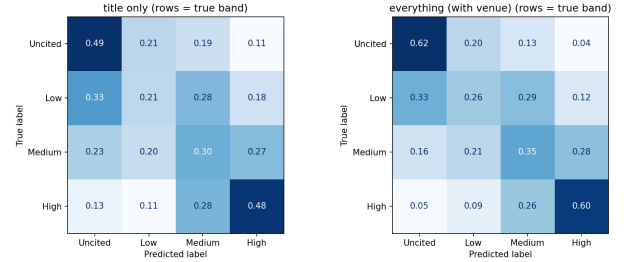


Fig. 2. Test-set confusion matrices, row-normalized by true band. The diagonal is correct; the corners are the most severe mistakes. Adding metadata and venue sharpens the Uncited and High bands and shrinks the worst-case errors.

macro-F1, and venue alone reaches 0.390. Adding venue to title and metadata lifts the score from 0.420 to 0.450.

3) *A larger model does not raise the ceiling.* On the validation split, a Naive Bayes classifier (MultinomialNB) and a linear SVM trained on titles stayed within 0.01 of logistic regression, and Random Forest and HistGradientBoosting did not outperform it either (metadata-only macro-F1: 0.429 logistic regression, 0.403 Random Forest, 0.422 HistGradientBoosting). The ceiling of roughly 0.45 appears to come from the available features rather than model capacity, consistent with what the original CS229 project suspected but did not test.

4) *The middle bands are hardest to separate.* With all features enabled, 62% of uncited papers and 60% of High-band papers land in the correct band, and the most severe mistakes (an Uncited paper predicted High, or vice versa) fall to 4–5%, down from 11–13% with the title alone (Fig. 2). The Low band (1–2 citations) is difficult in every configuration, with test F1 between 0.20 and 0.29.

VI. DISCUSSION AND LIMITATIONS

A paper’s title alone says relatively little about how often it will be cited. Its venue and the length of its reference list say more, and the best achievable score using only information available at publication time is about 0.45 macro-F1, clearly above the majority-class baseline, but far from precise. The demo reflects this directly: given only a title, it achieves 0.36 macro-F1.

Several limitations bound these conclusions. We used a stratified sample of 44,000 papers rather than the full OpenAlex CS corpus. Our main comparison uses a single model family; Random Forest and HistGradientBoosting were evaluated only on the validation split, and HistGradientBoosting only on venue and metadata features, since the full TF-IDF title features were too high-dimensional to densify. We did not evaluate neural or transformer-based text encoders. The reference-count feature likely proxies for paper type (surveys cite heavily) rather than causing citations, and should not be interpreted causally. The venue field has 9,658 distinct values (7,869 in the training split), which limits what simple one-hot encoding can capture. Abstracts, author-level features, and learned venue embeddings are natural next steps.

VII. CONCLUSION

We revisited the question of whether model capacity or feature availability bounds citation-band prediction from a paper’s title. Using a controlled ablation over title, metadata and venue features on 43,997 OpenAlex computer-science papers, we show that the approximately 0.45 macro-F1 ceiling observed in prior work is a feature limitation, not a model-capacity limitation: stronger classifiers do not surpass it, while adding metadata and venue information does. Title text alone remains a weak but non-trivial signal.

CODE AND DATA AVAILABILITY

Code: <https://github.com/Tharun-2605/citation-classification>

Data: <https://www.kaggle.com/datasets/tharunganesh172/citations-openalex-cs-2019-2023>

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